

Eyler integrallari

1. Beta funksiya

2. Gamma funksiya

3. Beta va gamma funksiyalar orasidagi bog'lanish

Biz 48-ma'ruzada ushbu

$$\int_0^1 x^{a-1}(1-x)^{b-1} dx$$

xosmas integralning $a>0, b>0$ bo'lganda yaqinlashuvchiligini,

$$\int_0^{+\infty} x^{a-1} e^{-x} dx$$

xosmas integralning esa $a>0$ bo'lganda yaqinlashuvchiligini isbotlagan edik.

Ravshanki, bu xosmas integrallar a va b larga bog'liq, ya'ni parametrga bog'liq xosmas integrallar bo'ladi.

1°. Beta funksiya va uning tekis yaqinlashuvchiligi. Ushbu

$$\int_0^1 x^{a-1}(1-x)^{b-1} dx$$

parametrga bog'liq xosmas integral beta funksiya (I -tur Eyler integrali) deyiladi va $B(a, b)$ kabi belgilanadi:

$$B(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx \quad (a>0, b>0).$$

Demak, beta funksiya

$$\{(a, b) \in \mathbb{R}^2 : a \in (0, +\infty), b \in (0, +\infty)\}$$

to'plamda aniqlangan funksiya.

1-teorema. Ushbu

$$B(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx$$

integral $M_0 = \{(a, b) \in \mathbb{R}^2 : a \in [a_0, +\infty), b \in [b_0, +\infty), a_0>0, b_0>0\}$ to'plamda tekis yaqinlashuvchi bo'ladi.

◀ $B(a, b)$ funksiyani ifodalovchi integralni ikki qismga

$$\int_0^1 x^{a-1}(1-x)^{b-1} dx = \int_0^{\frac{1}{2}} x^{a-1}(1-x)^{b-1} dx + \int_{\frac{1}{2}}^1 x^{a-1}(1-x)^{b-1} dx$$

ajratib, har bir integralning tekis yaqinlashishga tekshiramiz. Parametr $a \geq a_0$ ($a_0 > 0$), $\forall b > 0$,

$\forall x \in \left(0, \frac{1}{2}\right]$ da

$$x^{a-1}(1-x)^{b-1} \leq x^{a_0-1}(1-x)^{b-1} \leq 2x^{a_0-1}$$

va $a > 0$ bo'lganda

$$\int_0^{\frac{1}{2}} x^{a-1} dx$$

integralning yaqinlashuvchi bo'lishidan Veyershtass alomatiga ko'ra

$$\int_0^{\frac{1}{2}} x^{a-1}(1-x)^{b-1} dx$$

integralning $a \geq a_0$, ($a_0 > 0$) da tekis yaqinlashuvchiligini topamiz. Shuningdek, parametr $b \geq b_0$

($b_0 > 0$), $\forall a > 0$, $\forall x \in \left[\frac{1}{2}, 1\right)$ da

$$x^{a-1}(1-x)^{b-1} \leq x^{a-1}(1-x)^{b_0-1} \leq 2(1-x)^{b_0-1}$$

va $b > 0$ bo'lganda

$$\int_{\frac{1}{2}}^1 (1-x)^{b-1} dx$$

integralning yaqinlashuvchi bo'lishidan Veyershtass alomati-ga ko'ra

$$\int_{\frac{1}{2}}^1 x^{a-1}(1-x)^{b-1} dx$$

integralning $b \geq b_0$ ($b_0 > 0$) da tekis yaqinlashuvchi bo'lishini topamiz. Demak,

$$B(a, b) = \int_0^1 x^{a-1}(1-x)^{b-1} dx$$

integral

$$M_0 = \{(a, b) \in R^2 : a \in [a_0, +\infty), b \in [b_0, +\infty), a_0 > 0, b_0 > 0\}$$

to'plamda tekis yaqinlashuvchi bo'ladi. ►

Natija. $B(a, b)$ funksiya

$$M = \{(a, b) \in R^2 : a \in (0, +\infty), b \in (0, +\infty)\}$$

to'plamda uzluksiz bo'ladi.

◀ Bu tasdiq

$$\int_0^1 x^{a-1} (1-x)^{b-1} dx$$

integralning tekis yaqinlashuvchiligi hamda integral ostidagi funksiyaning M to'plamda uzluksiz bo'lishidan kelib chiqadi. ▶

2°. $B(a, b)$ funksiyaning xossalari. Endi

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx$$

funksiyaning xossalarini keltiramiz.

1) $B(a, b)$ funksiya a va b argumentlariga nisbatan simmetrik funksiya, ya'ni,

$$B(a, b) = B(b, a) \quad (a > 0, b > 0)$$

bo'ladi.

◀ $B(a, b)$ ni ifodalovchi integralda $x = 1 - t$ almashtirish bajarib topamiz:

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = - \int_1^0 (1-t)^{a-1} t^{b-1} dt = \int_0^1 t^{b-1} (1-t)^{a-1} dt = B(b, a) \quad . \quad \blacktriangleright$$

2) $B(a, b)$ funksiya quyidagicha ham ifoda qilinadi:

$$B(a, b) = \int_0^{+\infty} \frac{t^{a-1}}{(1+t)^{a+b}} dt \quad (1)$$

◀ $B(a, b)$ ni ifodalovchi integralda $x = \frac{t}{1+t}$ almashtirish bajarib topamiz:

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = \int_0^{+\infty} \left(\frac{t}{1+t} \right)^{a-1} \left(1 - \frac{t}{1+t} \right)^{b-1} \frac{dt}{(1+t)^2} = \int_0^{+\infty} \frac{t^{a-1}}{(1+t)^{a+b}} dt \quad \blacktriangleright$$

Agar (1) da $b = 1 - a$ ($0 < a < 1$) deyilsa, unda

$$B(a, 1-a) = \int_0^{+\infty} \frac{t^{a-1}}{1+t} dt = \frac{\pi}{\sin \pi a}$$

bo'ladi. Xususan,

$$B\left(\frac{1}{2}, \frac{1}{2}\right) = \pi$$

bo'ladi.

3) $B(a, b)$ funksiya uchun ushbu

$$B(a+1, b) = \frac{a}{a+b} B(a, b) \quad (a > 0, b > 0)$$

formula o'rinli bo'ladi.

◀ Ravshanki,

$$B(a+1, b) = \int_0^1 x^a (1-x)^{b-1} dx$$

Bu integralni bo'laklab integrallaymiz:

$$\begin{aligned} B(a+1, b) &= \int_0^1 x^a (1-x)^{b-1} dx = -\frac{1}{b} \int_0^1 x^a d((1-x)^b) = -\frac{1}{b} x^a (1-x)^b \Big|_0^1 + \frac{a}{b} \int_0^1 x^{a-1} (1-x)^b dx = \\ &= \frac{a}{b} \int_0^1 x^{a-1} (1-x)^b dx = \frac{a}{b} \left[\int_0^1 x^{a-1} (1-x)^{b-1} dx - \int_0^1 x^a (1-x)^{b-1} dx \right] = \frac{a}{b} B(a, b) - \frac{a}{b} B(a+1, b). \end{aligned}$$

Natijada

$$B(a+1, b) = \frac{a}{b} B(a, b) - \frac{a}{b} B(a+1, b) \quad (2)$$

bo'lib, undan

$$B(a+1, b) = \frac{a}{a+b} B(a, b)$$

bo'lishi kelib chiqadi. ▶

$B(a, b)$ funksiya simmetrik bo'lganligidan ushbu

$$B(a, b+1) = \frac{b}{a+b} B(a, b) \quad (3)$$

bo'ladi.

Natija. $B(m, n)$ funksiya ($m \in \mathbb{N}, n \in \mathbb{N}$) (2) va (3) formulalarni takror qo'llash natijasida

$$B(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!}$$

bo'lishi kelib chiqadi.

3°. Gamma funksiya va uning yaqinlashuvchiligi. Ushbu

$$\int_0^{+\infty} x^{a-1} e^{-x} dx$$

parametr ga bog'liq xosmas integral gamma funktsiya (II-tur Eyler integrali) deyiladi va $\Gamma(a)$ kabi belgilanadi:

$$\Gamma(a) = \int_0^{+\infty} x^{a-1} e^{-x} dx.$$

Demak, gamma funktsiya $(0, +\infty)$ da aniqlangan funktsiya.

2-teorema. Ushbu

$$\Gamma(a) = \int_0^{+\infty} x^{a-1} e^{-x} dx$$

integral $[a_0, b_0]$ da $(0 < a_0 < b_0 < +\infty)$ tekis yaqinlashuvchi bo'ladi.

◀ $\Gamma(a)$ funktsiyani ifodalovchi integralni ikki integral yig'indisi sifatida yozib olamiz:

$$\Gamma(a) = \int_0^1 x^{a-1} e^{-x} dx + \int_1^{+\infty} x^{a-1} e^{-x} dx.$$

So'ng ikkala integralning ixtiyoriy $[a_0, b_0]$ segmentda $(0 < a_0 < b_0 < +\infty)$ tekis yaqinlashuvchi bo'lishini ko'rsatamiz. Parametr $a \geq a_0$ ($a_0 > 0$), $\forall x \in (0, 1]$ da

$$x^{a-1} e^{-x} \leq x^{a_0-1}$$

va $a_0 > 0$ da

$$\int_0^1 x^{a_0-1} dx$$

integralning yaqinlashuvchi bo'lishidan Veyershtass alomati-ga ko'ra

$$\int_0^1 x^{a-1} e^{-x} dx$$

integralning $a \geq a_0$ da tekis yaqinlashuvchi bo'lishi kelib chiqadi. Shuningdek, parametr $a \leq b_0$, $\forall x \in [1, +\infty)$ da

$$x^{a-1} e^{-x} \leq x^{b_0-1} e^{-x}$$

va

$$\int_1^{+\infty} x^{b_0-1} e^{-x} dx$$

integralning yaqinlashuvchi bo'lishidan yana Veyershtrass alomatiga ko'ra

$$\int_1^{+\infty} x^{a-1} e^{-x} dx$$

integralning $a \leq b_0$ da tekis yaqinlashuvchi bo'lishini topamiz. Demak,

$$\Gamma(a) = \int_0^{+\infty} x^{a-1} e^{-x} dx$$

xosmas integral $[a_0, b_0]$ da tekis yaqinlashuvchi bo'ladi. ►

Natija. $\Gamma(a)$ funksiya $(0, +\infty)$ da uzluksiz bo'ladi.

◀ Bu tasdiq

$$\int_0^{+\infty} x^{a-1} e^{-x} dx$$

integralning tekis yaqinlashuvchiligi hamda integral ostidagi funksiyaning $M = \{(x, a) \in R^2 : x \in (0, +\infty), a \in (0, +\infty)\}$ da uzluksiz bo'lishidan kelib chiqadi. ►

4^o. $\Gamma(a)$ funksiyaning xossalari. 1) Gamma funksiya $(0, +\infty)$ da barcha tartibdagi uzluksiz hosilalarga ega va

$$\Gamma^{(n)}(a) = \int_0^{+\infty} x^{a-1} e^{-x} (\ln x)^n dx \quad (n=1, 2, 3, \dots)$$

bo'ladi.

◀ Ravshanki,

$$\Gamma(a) = \int_0^{+\infty} x^{a-1} e^{-x} dx$$

integral ostidagi $f(x, a) = x^{a-1} e^{-x}$ funksiya

$$M = \{(x, a) \in R^2 : x \in (0, +\infty), a \in (0, +\infty)\}$$

to'plamda uzluksiz bo'lib, uzluksiz

$$f'_a(x, a) = x^{a-1} e^{-x} \ln x$$

hosilaga ega bo'ladi. Yuqorida aytganimizdek

$$\Gamma(a) = \int_0^1 x^{a-1} e^{-x} dx + \int_1^{+\infty} x^{a-1} e^{-x} dx$$

tenglikning o'ng tomonidagi integrallar ixtiyoriy $[a_0, b_0]$ da $(0 < a_0 < b_0 < +\infty)$ tekis yaqinlashuvchi.

Ushbu $\int_0^1 x^{a-1} \ln x \cdot e^{-x} dx$, $\int_1^{+\infty} x^{a-1} \ln x \cdot e^{-x} dx$ integrallarni qaray-lik. Bu integrallardan birinchisi, $a \geq a_0 > 0$ da

$$|x^{a-1} \ln x \cdot e^{-x}| \leq x^{a_0-1} |\ln x|$$

va

$$\int_0^1 x^{a_0-1} |\ln x| dx$$

integral yaqinlashuvchi bo'lganligidan Veyersstrass alomatiga ko'ra tekis yaqinlashuvchi bo'ladi. Shuningdek ikkinchi integral ham, $a \leq b_0 < +\infty$ da

$$|x^{a-1} \ln x \cdot e^{-x}| \leq x^{b_0-1} \ln x \cdot e^{-x} = x^{b_0-1} e^{-x}$$

va

$$\int_1^{+\infty} x^{b_0-1} \ln x \cdot e^{-x} dx$$

integral yaqinlashuvchi bo'lganligidan yana Veyersstrass alomatiga ko'ra tekis yaqinlashuvchi bo'ladi. Parametrga bog'liq xosmas integralning parametr bo'yicha differensiallash haqidagi teoremdan foydalanib topamiz:

$$\begin{aligned} \frac{d}{da} \Gamma(a) &= \frac{d}{da} \left[\int_0^1 x^{a-1} e^{-x} dx + \int_1^{+\infty} x^{a-1} e^{-x} dx \right] = \int_0^1 \frac{d}{da} (x^{a-1} e^{-x}) dx + \\ &+ \int_1^{+\infty} \frac{d}{da} (x^{a-1} e^{-x}) dx = \int_0^1 x^{a-1} \cdot \ln x \cdot e^{-x} dx + \int_1^{+\infty} x^{a-1} \cdot \ln x \cdot e^{-x} dx. \end{aligned}$$

Demak,

$$\Gamma'(a) = \int_0^{+\infty} x^{a-1} \cdot \ln x \cdot e^{-x} dx$$

$\Gamma'(a)$ funksiyaning $[a_0, b_0]$ da uzluksiz bo'lishi ravshan.

Xuddi shu yo'l bilan $\Gamma(a)$ funksiyaning ikkinchi, uchinchi va hokazo tartibdagi hosilalarining mavjudligi, uzluksizligi hamda

$$\Gamma^{(n)}(a) = \int_0^{+\infty} x^{a-1} e^{-x} (\ln x)^n dx$$

bo'lishi ko'rsatiladi. ►

2) $\Gamma(a)$ funksiya uchun ushbu

$$\Gamma(a+1) = a\Gamma(a) \quad (4)$$

formula o'rinli bo'ladi.

◄Ravshanki,

$$\Gamma(a) = \int_0^{+\infty} x^{a-1} e^{-x} dx$$

Bu integralni bo'laklab integrallaymiz. Natijada

$$\Gamma(a+1) = \int_0^{+\infty} x^a e^{-x} dx = - \int_0^{+\infty} x^a d(e^{-x}) = -x^a e^{-x} \Big|_0^{+\infty} + a \int_0^{+\infty} x^{a-1} e^{-x} dx = a\Gamma(a)$$

bo'ladi. ►

Ma'lumki, $a \in (0, 1]$ bo'lsa, $a+1 \in (1, 2]$ bo'ladi. $\Gamma(a)$ funksiyaning bu xossasini ifodalovchi (4) munosabat $\Gamma(a)$ funksiyaning $(0, 1]$ dagi qiymatlariga ko'ra uning $(1, 2]$ oraliqdagi qiymatlarini, umuman ixtiyoriy $(n, n+1]$ dagi qiymatlarini topish imkonini beradi.

Natija. $\Gamma(n)$ funksiya ($n \in \mathbb{N}$) (4) formulani takror qo'llash natijasida ($\Gamma(1) = 1$)

$$\Gamma(n) = (n-1)!$$

bo'lishi kelib chiqadi.

3) $\Gamma(a)$ funksiyaning o'zgarish xarakteri. Ravshanki,

$$\Gamma(1) = \int_0^{+\infty} e^{-x} dx = 1$$

Yuqoridagi (4) formulaga ko'ra

$$\Gamma(2) = 1 \cdot \Gamma(1) = 1$$

bo'ladi. Roll teoremasiga muvofiq, shunday a_0 ($1 < a_0 < 2$) nuqta topiladiki,

$$\Gamma'(a_0) = 0$$

bo'ladi. Ayni paytda, $\forall a \in (0, +\infty)$ da

$$\Gamma''(a) = \int_0^{+\infty} x^{a-1} e^{-x} \ln^2 x dx > 0$$

bo'lganligi uchun $\Gamma'(a)$ funksiya $(0, +\infty)$ da qat'iy o'suvchi bo'ladi. Binobarin, $\Gamma'(a)$ funksiya a_0 nuqtadan boshqa nuqtalarda nolga aylanmaydi. Demak,

$$\Gamma'(a) = \int_0^{+\infty} x^{a-1} e^{-x} \ln x dx = 0$$

tenglama $(0, +\infty)$ oraliqda yagona yechimga ega. Unda,

$$0 < a < a_0 \text{ da } \Gamma'(a) < 0,$$

$$a_0 < a < +\infty \text{ da } \Gamma'(a) > 0$$

bo'lib, $\Gamma(a)$ funksiya a_0 nuqtada minimumga ega bo'ladi. ($a_0 = 1,4616\dots$, $\Gamma(a_0) = \min \Gamma(a) = 0,8856\dots$ bo'lishi topilgan).

$\Gamma(a)$ funksiya $a > a_0$ da o'suvchi bo'lganligi sababli $a > n+1$ bo'lganda $\Gamma(a) > \Gamma(n+1) = n!$ bo'lib, undan

$$\lim_{a \rightarrow +\infty} \Gamma(a) = +\infty$$

bo'lishi kelib chiqadi.

Agar $a \rightarrow +0$ da $\Gamma(a+1) \rightarrow \Gamma(1) = 1$ hamda

$$\Gamma(a) = \frac{\Gamma(a+1)}{a}$$

bo'lishini e'tiborga olsak, unda

$$\lim_{a \rightarrow +0} \Gamma(a) = +\infty$$

ekanligini topamiz.

5°. Beta va gamma funksiyalar orasidagi bog'lanish. Beta va gamma funksiyalar orasidagi bog'lanishni quyidagi teorema ifodalaydi.

3-teorema. $\forall (a, b) \in \{(a, b) \in \mathbb{R}^2 : a \in (0, +\infty), b \in (0, +\infty)\}$ uchun

$$B(a, b) = \frac{\Gamma(a) \cdot \Gamma(b)}{\Gamma(a+b)} \quad (5)$$

formula o'rinli bo'ladi.

◀ Ushbu

$$\Gamma(s) = \int_0^{+\infty} x^{s-1} e^{-x} dx$$

integralda $x = (1+u)t$, ($t > 0$) almashtirish bajarib, s ni $a+b$ ga almashtiramiz. Natijada

$$\Gamma(a+b) = \int_0^{+\infty} (1+u)^{a+b-1} t^{a+b-1} e^{-(1+u)t} (1+u) dt$$

bo'lib,

$$\frac{\Gamma(a+b)}{(1+u)^{a+b}} = \int_0^{+\infty} t^{a+b-1} e^{-(1+u)t} dt$$

bo'ladi.

Endi bu tenglikning har ikki tomonini u^{a-1} ga ko'paytirib, so'ng $(0, +\infty)$ oraliq bo'yicha integrallab topamiz:

$$\Gamma(a+b) \int_0^{+\infty} \frac{u^{a-1}}{(1+u)^{a+b}} du = \int_0^{+\infty} \left[\int_0^{+\infty} t^{a+b-1} e^{-(1+u)t} dt \right] u^{a-1} du$$

ya'ni,

$$\Gamma(a+b) \cdot B(a, b) = \int_0^{+\infty} \left[\int_0^{+\infty} t^{a+b-1} e^{-(1+u)t} dt \right] u^{a-1} du$$

[1], 17-bob, 8-§da keltirilgan teoremdan foydalanib, keyingi tenglikning o'ng tomonidagi integrallarning o'rinlarini almashtiramiz. Natijada

$$\Gamma(a+b) \cdot B(a, b) = \int_0^{+\infty} \left[\int_0^{+\infty} u^{a-1} e^{-(1+u)t} du \right] t^{a+b-1} dt$$

bo'ladi. Integralda $ut = y$ almashtirish bajarib topamiz:

$$\Gamma(a+b) \cdot B(a, b) = \int_0^{+\infty} \left[\int_0^{+\infty} y^{a-1} t^{b-1} e^{-t} e^{-y} dy \right] dt = \int_0^{+\infty} t^{b-1} e^{-t} dt \cdot \int_0^{+\infty} y^{a-1} e^{-y} dy = \Gamma(b) \cdot \Gamma(a).$$

Demak,

$$B(a, b) = \frac{\Gamma(a) \cdot \Gamma(b)}{\Gamma(a+b)}.$$

Natija. $\forall a \in (0, 1)$ uchun

$$\Gamma(a) \Gamma(1-a) = \frac{\pi}{\sin \pi a}$$

bo'ladi.

◀ (5) tenglikda $b = 1 - a$ ($0 < a < 1$) deb olinsa, unda

$$B(a, 1-a) = \frac{\Gamma(a) \cdot \Gamma(1-a)}{\Gamma(1)}$$

bo'ladi. Ma'lumki,

$$B(a, 1-a) = \frac{\pi}{\sin a\pi}, \Gamma(1) = 1.$$

Demak,

$$\Gamma(a) \Gamma(1-a) = \frac{\pi}{\sin a\pi}, (0 < a < 1). \blacktriangleright$$

Agar (6) formulada $a = \frac{1}{2}$ deyilsa, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ bo'lishi kelib chiqadi.

1-misol. 1. Ushbu

$$\int_0^{+\infty} e^{-x^2} dx$$

integral hisoblansin.

◀ Bu integralda $x^2 = t$ almashtirish bajaramiz. Unda

$$dx = \frac{1}{2\sqrt{t}} dt = \frac{1}{2} t^{-\frac{1}{2}} dt$$

bo'lib,

$$\int_0^{+\infty} e^{-x^2} dx = \frac{1}{2} \int_0^{+\infty} t^{-\frac{1}{2}} e^{-t} dt = \frac{1}{2} \int_0^{+\infty} t^{\frac{1}{2}-1} e^{-t} dt = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$$

bo'ladi. ▶

2-misol. Ushbu $\int_0^{+\infty} \frac{dx}{1+x^3}$ integral hisoblansin.

$$1+x^3 = \frac{1}{y}$$

◀ Bu integralda y almashtirish bajaramiz. Unda

$$x = \left(\frac{1-y}{y}\right)^{\frac{1}{3}}, dx = \frac{1}{3} \left(\frac{1-y}{y}\right)^{-\frac{2}{3}} \cdot \left(-\frac{dy}{y^2}\right),$$

$$\int_0^{+\infty} \frac{dx}{1+x^3} = -\frac{1}{3} \int_1^0 y^{\frac{1}{3}} \left(\frac{1-y}{y} \right)^{-\frac{2}{3}} \cdot \frac{dy}{y^2} = \frac{1}{3} \int_0^1 y^{-\frac{1}{3}} (1-y)^{-\frac{2}{3}} dy =$$

$$\frac{1}{3} B\left(\frac{2}{3}, \frac{1}{3}\right) = \frac{1}{3} \cdot \frac{\Gamma\left(\frac{1}{3}\right) \cdot \Gamma\left(\frac{2}{3}\right)}{\Gamma(1)} = \frac{1}{3} \cdot \frac{\pi}{\sin \frac{1}{3}\pi} = \frac{\pi}{3 \cdot \frac{\sqrt{3}}{2}} = \frac{2\pi}{3\sqrt{3}}$$

bo'ladi. ►

Mashqlar

1. $\forall a \in R(0, -1, \dots, -n, \dots)$ da $\Gamma(a) = \lim_{n \rightarrow \infty} \frac{n^a \cdot n!}{a(a+1) \dots (a+n)}$ bo'lishi isbotlansin.
2. Ushbu $\int_0^{+\infty} \frac{x^m dx}{(a+bx^n)^p}$ ($a>0, b>0, n>0$) integral beta funksiya orqali ifodalansin.
3. Quyidagi $\left(\Gamma'(a)\right)^2 < \Gamma(a) \cdot \Gamma''(a)$ tengsizlik isbotlansin.